

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Doctor Appointment Booking System

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Doctor Appointment Booking System**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Preeti Saini**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 08.05.2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude and heartfelt thanks to our project guide, Dr. Preeti Saini, Assistant Professor, Department of Computer Science and Engineering, Chitkara University Institute of Engineering and Technology, for her valuable guidance, constant encouragement, and continuous support throughout the development of our project titled **“Doctor Appointment Booking System.”** Her expert suggestions, constructive feedback, and motivation helped us successfully complete this project work.

We are also thankful to all the faculty members and staff of the Department of Computer Science and Engineering for providing us with the necessary facilities and support required during the course of this project.

We would also like to thank our parents and friends for their encouragement, cooperation, and moral support throughout the project duration.

Lastly, we express our sincere gratitude to Chitkara University for providing us with the opportunity to work on this project and enhance our technical knowledge and practical skills.

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ABSTRACT

The Doctor Appointment Booking System is a web-based healthcare management application developed using the MERN Stack, which includes MongoDB, Express.js, React.js, and Node.js. The main objective of the system is to simplify and digitize the traditional process of booking doctor appointments by providing an efficient, secure, and user-friendly online platform for patients, doctors, and administrators.

In traditional healthcare systems, appointment scheduling is generally performed manually, which often results in long waiting times, appointment conflicts, paperwork, and inefficient communication between patients and healthcare providers. Managing appointments manually also increases the workload on hospital staff and creates inconvenience for patients. The proposed system addresses these challenges by automating the appointment management process and enabling users to access healthcare services digitally.

The platform allows patients to register and log in securely, search for doctors based on specialization, experience, and availability, and book appointments online without physically visiting hospitals or clinics. Doctors can manage their schedules, update appointment availability, and monitor patient appointments through a dedicated dashboard. Administrators are provided with complete control over doctor management, appointment monitoring, and overall system administration.

The application is developed using React.js for creating a responsive and interactive frontend user interface, while Node.js and Express.js are used for backend development and API handling. MongoDB is used as the database management system for storing user information, doctor details, appointment records, and payment data securely. JWT (JSON Web Token) authentication is implemented to provide secure login functionality and role-based access control for different users of the system.

The project also supports online payment integration using Razorpay, allowing patients to make secure digital payments for appointments and consultations. RESTful APIs are used for efficient communication between the frontend and backend components of the application.

INTRODUCTION

The Doctor Appointment Booking System is a web-based healthcare management application developed using the MERN Stack, which includes MongoDB, Express.js, React.js, and Node.js. The project aims to digitize and simplify the traditional process of booking doctor appointments by providing an online platform for patients, doctors, and administrators.

In traditional healthcare systems, patients often face difficulties such as long waiting times, manual appointment scheduling, paperwork, and lack of proper communication between doctors and patients. Managing appointments manually also creates scheduling conflicts and inefficiencies for hospitals and clinics. To overcome these challenges, the Doctor Appointment Booking System provides an automated and user-friendly solution.

The system allows patients to register and log in securely, search for doctors based on specialization, experience, and availability, and book appointments online without visiting hospitals physically. Doctors can manage their appointment schedules, update availability, and monitor patient bookings through a dedicated dashboard. Administrators have complete control over user management, doctor approvals, and appointment monitoring.

The project focuses on improving healthcare accessibility, reducing manual work, and enhancing user experience through a responsive and secure platform. JWT authentication is implemented to ensure secure login and role-based access control for patients, doctors, and administrators.

The platform is designed to be scalable, efficient, and user-friendly, making it suitable for hospitals, clinics, and healthcare organizations seeking digital transformation.

METHODOLOGY

The development of the Doctor Appointment Booking System followed a structured software development methodology consisting of planning, requirement analysis, system design, development, testing, and deployment phases. Agile methodology was adopted during development to ensure flexibility, continuous improvement, and effective project management.

Requirement Analysis

The first step involved understanding the requirements of patients, doctors, and administrators. The major requirements identified were:

- Online appointment booking functionality
- Secure user authentication
- Doctor profile management
- Real-time appointment scheduling
- Admin monitoring and control
- Responsive user interface

System Design

After requirement analysis, the system architecture and database schema were designed. The frontend and backend modules were separated to maintain scalability and modularity. MongoDB collections were designed for storing:

- User information
- Doctor details
- Appointment records
- Authentication data

Tools and Technologies

The Doctor Appointment Booking System was developed using modern web development technologies to ensure scalability, security, and performance.

Technology	Purpose
React.js	Frontend User Interface Development
Node.js	Backend Runtime Environment
Express.js	Server and API Development
MongoDB	Database Management
JWT Authentication	Secure Authentication
CSS / Bootstrap / Tailwind CSS	Styling and Responsive Design
Postman	API Testing
Git & GitHub	Version Control and Collaboration
VS Code	Code Editor
Razorpay	Online Payment Integration

React.js

React.js is used to create dynamic and responsive user interfaces. It allows component-based development, improving code reusability and maintainability.

Node.js and Express.js

Node.js provides a runtime environment for executing JavaScript on the server side. Express.js simplifies backend development by providing routing and middleware support.

MongoDB

MongoDB is a NoSQL database used to store patient records, doctor information, and appointment details. It provides flexibility and scalability for handling large amounts of healthcare data.

Razorpay Integration

Razorpay payment gateway integration allows patients to make secure online payments for appointments and consultations.

Implementation

The implementation phase involved integrating frontend, backend, and database modules to create a complete healthcare appointment management platform.

User Registration and Login

Patients, doctors, and administrators can register and log in securely. JWT authentication ensures secure access and role-based authorization.

Doctor Management Module

Doctors can:

- Create profiles
- Update specialization and experience
- Set appointment availability
- View patient appointments

Administrators can approve or manage doctor accounts through the admin dashboard.

Appointment Booking Module

Patients can:

- Search doctors by specialization
- View available slots
- Book appointments online
- Track appointment status

Appointment details are stored securely in MongoDB.

Admin Dashboard

The admin dashboard provides:

- User management

- Doctor approval system
- Appointment monitoring
- System control and analytics

Payment Integration

Razorpay payment gateway integration enables secure online payments for appointments. Payment details are verified before confirming bookings.

Database Integration

MongoDB collections are used for:

- User Data
- Doctor Profiles
- Appointment Records
- Payment Information

API Integration

REST APIs were implemented for:

- Authentication
- Appointment management
- Doctor management
- Payment processing

The frontend communicates with backend APIs using Axios for data exchange and real-time updates.

Major Findings / Outcomes / Results

The Doctor Appointment Booking System successfully automates healthcare appointment management and improves the efficiency of hospital and clinic operations.

Major Findings

Improved Appointment Scheduling

The system eliminates manual appointment booking and allows patients to schedule appointments quickly and efficiently.

Reduced Waiting Time

Patients can book appointments remotely without standing in long queues at hospitals or clinics.

Secure Data Management

JWT authentication and encrypted passwords improve data security and protect sensitive user information.

Better User Experience

The responsive interface and organized dashboards provide a smooth and user-friendly experience for patients and doctors.

Efficient Doctor Management

Doctors can manage schedules and appointments easily through personalized dashboards.

Scalable Architecture

The MERN Stack architecture ensures scalability and flexibility for future enhancements and large-scale deployment.

Conclusion and Future Scope

The Doctor Appointment Booking System demonstrates the practical implementation of MERN Stack technologies in healthcare management. The project successfully addresses major issues associated with traditional appointment systems such as manual scheduling, long waiting times, and inefficient management processes.

The platform provides secure authentication, real-time appointment booking, doctor management, and responsive dashboards for patients, doctors, and administrators. By digitizing healthcare appointment management, the system improves operational efficiency and enhances the overall patient experience.

The project also demonstrates the effectiveness of modern web technologies such as React.js, Node.js, Express.js, MongoDB, and JWT authentication in building scalable and secure applications.

Future Scope

The project can be enhanced further with advanced features such as:

Video Consultation

Integration of real-time video consultation functionality for remote healthcare services.

AI-Based Doctor Recommendation

Artificial Intelligence can recommend doctors based on symptoms and patient history.

Cloud Deployment and Scalability

Cloud hosting can improve scalability, performance, and availability for large healthcare organizations.

REFERENCES

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- [React.js Official Documentation](#)
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- Database Management System Concepts and Web Development References
- API Development and RESTful Services Documentation

